

WORK PROCEDURE

PRO-OP-ITH-007A

STANDARD FOR V-30 BLIND BORING UP 30" DIAMETER

OBJECTIVE

The objective of this procedure is to define safe working methods for blind boring up with the drilling head V-30.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

Personnel with the authority for the interpretation of this document are the director general and operations director. Any request for amendment, revision or update must be made with one of these persons.

RESPONSIBILITIES

The worker responsible for the drilling must ensure to follow these steps and follow the instruction that come from the engineering department.

SAFETY PRECAUTIONS AND IMPORTANT INSTRUCTIONS.

- 1) MRI employees can move the drilling equipment into the workplace using the company's equipment provided all company regulations and directives are adhered to.
- 2) A visual inspection of the unit is completed once the unit arrives at the work place. Ventilation, power supply and air and water distribution will also be verified.
- 3) The electric cable is unloaded, inspected and connected to the ground fault distribution box. (an electrician from the mine is required just in case a motor rotation re-connect is required (ex.: inverse rotation))

- 4) Verification of all control levers is required prior to moving the unit. Control levers must be in the neutral (deadman's) position before powering up the unit
- 5) A visual inspection of the unit is required for oil levels, gauges, leaks, etc. prior to moving the unit.

 INSPECTIONS

It is essential to carry out a rigorous inspection of our equipment as soon as we have to operate it. In the case of CUBEX and boosters, particular attention must be paid to the high-temperature (800 degrees F) braided hoses of the compressors. For safety reasons, only the type of hose as shown below should be used.



- 6) Before the unit is moved to the first line to be drilled, with respect to the plans and specifications, a wash down of the area, a check scale and an inspection for explosives is performed in the drilling area.
- 7) Work may be interrupted at any time to scale or purge and secure the protective cage.

- 8) When the drill is moved in the drilling area, we take precautions to avoid damage to the electrical cable connected to the drill and check if there are no leaks or abnormalities from the drill in motion. If the 600V cable has to be handled when it is turned on, you must use the plastic hook.
- 9) Barricades and or signs are installed to advise other workers around of the ongoing operations (Drilling, Open Holes, etc.). A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.) See **INSTALLATION STANDARD**.
- 10) After reaching the drilling point, the drill is turned off with the **E-STOP** and the electric cable is then hung to the wall at chest height (4 to 5 feet off the bench) such as to eliminate any contact by loose or motor vehicles. Also, as the cable is being hung, a visual inspection of the cable is done.



The use of the E-Stop to stop the drill will result of cutting the 600 V current on all extensions.

- 11) The connection of the low-pressure air hose (2 ½ in. dia.) from the mine air supply to the drill's booster compressor is the next priority. The airline is blown clean and safety-restraining devices (whip checks) are installed at each connection and splice (if there happens to be one). At this point, an inspection of the retaining device is important. You must check the condition of the wire and the proper functioning of clamping mechanisms.
- 12) The re-enforced high-pressure hose (2 ½) is inspected, blown and connected from the compressor to the drill unit using whip checks at all the connections. (Note: Stem nuts on the 2 ½ in. dia. hoses are designed so that they cannot be interchanged between high and low pressure so as to eliminate, as much as possible, any chance of a potential accident

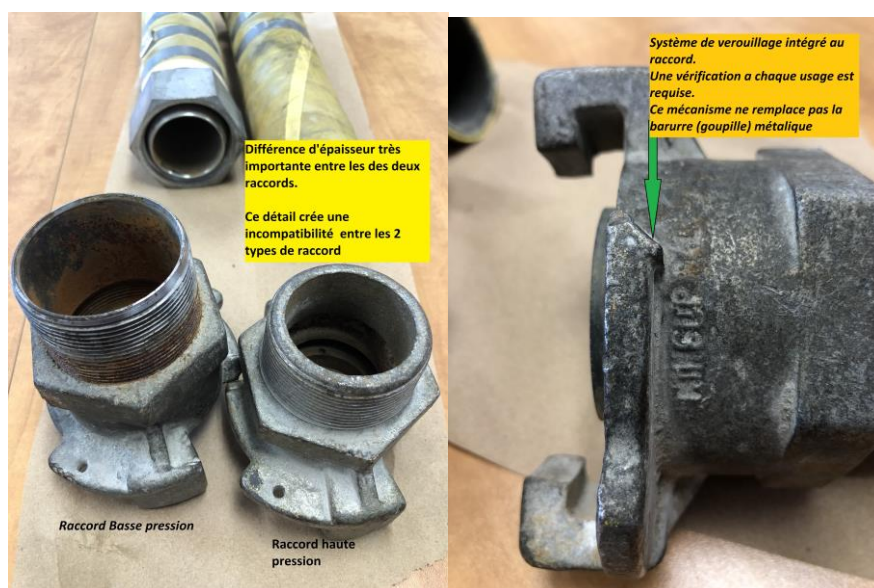


Special attention required

- 13) Connections and hoses should be thoroughly inspected to ensure that the collars are tight, that the threads of the couplings are in good condition, and that the hoses are not faulty**

Difference in thickness between the two connections. This detail creates an incompatibility between the two types of fittings

Locking system integrated into the fittings. A check for each use is required. This mechanism does not replace the metal pin



- 14)** The 1 in. dia. water hose is inspected, blown and connected from the mine supply to the drill.
- 15)** All hoses are hung under the electric cable to eliminate any contact with falling rocks or mine equipment.
- 16)** If it is not physically possible to hang the cable and hoses (ex. due to stope dimensions) they are laid out in such a fashion as to eliminate contact or crushing. They may be raised off the bench using drums or wooden tri-pods and marked (painted or identified with flagging) so that they will be visible.
- 17)** Air and water lines are then slowly energized and inspected for leaks.
- 18)** When moving the rotating mast, special attention must be paid to the hydraulic hoses, they could get stuck. If so, go back and move the mast

to loosen them. If an effort is necessary to loosen the hoses, make sure you are in a stable position.

- 19) For production holes greater than 6½ inch in diameter, the operator will require the assistance of a driller helper when handling the hammer. The hammer must be handled using the transport carriage. **Make sure that the transport carriage is stable before laying the hammer on.** Bits changes and rods handling can be handled by the operator only, depending on the length of holes to be drilled. Hole length of 200 feet and greater will require the assistance of a driller helper for rod handling, due to the physical effort required.
- 20) The installation of a deflector is required to redirect all drilling debris away from the employees and equipment.
- 21) During the drilling of the hole no one (employee or other) is authorized to circulate in front of the drilling unit.
- 22) To remove or install drill rods the gear box must be lowered and de-energized. The drill must be in the idle position before rods are manipulated.
- 23) Once the 6½ in. dia. pilot hole has been completed it must be reamed to 10½ in. in diameter for the V-30 guide sleeve. The same method for rod installation and removal applies.
- 24) Access between controls, drill mast and drill pipe support must be clear at all times (no hoses, no casing, no rocks).
- 25) As soon as the V-30 head is completely into the rock (approx. 6 ft) the unit is stopped and the deflector pads are reinstalled on the four sides of the raise so as to offer the maximum protection for the employees and the equipment.
- 26) The deflector pads are made with wire mesh, covered with canvas if necessary. They are suspended from the back with chains, ropes or shackles, and installed in such a way as to allow the panels to be opened and closed;

PROCEDURE

- 1- Before mobilizing the drill and all the equipment required for drilling, the height between the workfloor and the back must be between 12 feet minimum, and a maximum of 14 feet, depending on the degree of drilling. When the space between the workfloor and the back of the work area exceeds 4.3m (14ft), the workfloor area must be raised with muck:

- a. The height must allow the bits to be in the raise when the V-30 is in the slip plate **(note that the measurement between the slip plate and the end of the bits is 7 feet and 6 inches).**
 - b. For all v-30 up-hole drilling, there should be a maximum distance of 2 to 4 feet between the tables and the hole collar, which will allow better load distribution on our mechanical components.
- 2- When arriving at the site with the equipment, position it according to the INSTALLATION STANDARD (following the procedure);
 - 3- Make sure that the material or equipment is positioned in the stope in such a way as it's easy to access;
 - 4- Always maintain a distance of 7 meters between the drill and the compressor;
 - 5- Align the drill under the drilling target (in line and at the desired angle) according to the plans and specifications provided by engineering, using an inclinometer (degree hole) for quality control;
 - 6- The drilling control must be positioned 5 meters from the drilling point to be safe from rock projection;
 - 7- Drill the pilot hole 6 ½" and then 10".
 - 8- When the 10" is complete and the rods have all been removed from the hole, proceed with the V-30 installation.

INSTALLATION OF THE V-30 ON THE BOOM OF THE DRILL

- 1- Never stand in the axis of possible movement of the head if it is unhooked accidentally.**
 - 2- Make sure the controls are on neutral position**
1. Position the drill in the center of the V-30 at a distance of approximately 15 feet. The head should be oriented as follows; stabilizer towards the drill and the centralizer pointing forward
 2. Tilt the boom at an angle of about 60 degrees to the rear, keeping the base of the boom on the ground. If it is not possible to support the base on the ground to approach the head, extend the stinger to create a support and stabilize the mast and the drill.
 3. Install the male adapter in the gear box as well as the female adapter on the drive shaft of the V-30

4. Install the chain (grade 80, 12 feet) between the gear box of the drill and the drive shaft of the V-30.
5. Use the feed cylinder to hoist the V-30 to the table until the drive shaft has entered the table enough to couple the gear box to the drive shaft of the V-30
6. Close the centralizers on the drive shaft of the V-30
7. Shut the drill OFF and remove the adapters
8. Position the mast to align the drive shaft and the gear box
9. Approach the gear box on the drive shaft and start the rotation to screw them together.
10. Put the slip plate in the slot of the rod to tighten the assembly
11. Remove the pin and open the tables
12. Fully hoist the V-30 onto the boom
13. Close the tables on each side of the V-30 to stabilize it
14. Realign the boom into the initial drilling position.
15. If the raise planned to go beyond 40 feet, centralizers must be installed every 40 feet to stabilize the rods.

Cuttings

- Reaming the hole to 30" produces cuttings. If these cuttings interfere with drilling operations, stop drilling and blow these cuttings to the side using an air hose without exposing yourself directly under the V-30 hole. It is recommended to wear a mask when blowing drill cuttings.
- It is possible for rocks to end up on the tables before they can be closed.
 - i. If they are not causing harm, they can remain on the tables and drilling can continue until the tables can be closed;
 - ii. If a rock must be removed, the use of water is recommended;
 - iii. If the rock does not wash away with water,
 1. The driller must completely remove the V-30 from the hole,
 2. Move the drill away from the hole so as not to expose yourself,
 3. Move the mast to the side,
 4. Remove the rock
 5. Replace the drill and put the V-30 back in the hole
 6. Continue drilling

- The accumulation of drilling waste or rock on the deflectors and in the screens must be cleaned before exposing under the deflectors.
- If we need to sharpen the bits, use the **Procedure PRO-SEC-010A Bit sharpening**
- Once the 30 inches raise has been completed and the equipment has been moved out, a barricade or the necessary signs are posted so everyone is notified that there is an open hole condition. The company is informed of the situation, and they can undertake the necessary safety precautions required.

PROCEDURE FOR MAINTENANCE ON THE V-30 AND CHANGING BITS

- 1- Lower the V-30 until it is screwed onto the gear box;
- 2- Lay the mast on the drill in transport position;
- 3- Fold down the protective cage mesh to close the hole;
- 4- Thoroughly clean out any cutting residues in and around the tractions before moving the drill;
- 5- The drill must be moved so that during maintenance, the worker is :
 - a. **5 meters from the drilling line of the rig and,**
 - b. **Out of the line of fire of any object or rock that may fall from the raise;**
- 6- Always put the nose of the V-30 on the ground for maintenance.
- 7- When maintenance is complete:
 - a. Lay the mast on the drill in transport position;
 - b. Return the rig to the drilling line;
 - c. Return the V-30 in the raise;
 - d. Reposition the protective cage.

PROCEDURE FOR DEALING WITH BROKEN ROD IN THE HOLE:

- 1- Should a rod failure occur, immediately stop drilling, shut down electrical power and air supply to the drill.
- 2- A barricade or the necessary signs must be posted at the entrance to the work area in such a manner as to close off the work area and preventing all access to the site.

- 3- The driller must notify the situation to his supervisor immediately and no work will resume until the supervisor is on site.
- 4- The driller assisted by the site supervisor will assess the situation and determine how the failed rod or rods will be removed and replaced.
- 5- The site supervisor will remain on site throughout the recovery process.
- 6- All drill operation required to remediate the situation will be carried out using the remote-control consol. The remote control should be positioned:
 - a. **At a minimum of 5 meters from the drilling line of the raise;**
 - b. **Out of the line of fire of objects or rocks that may fall from the mount;**

A RISK ANALYSIS HAS TO BE DONE AT THIS POINT

- 7- Once the failed rods have been removed and replaced, all remaining rods in the hole will be removed to ensure no other rods in the drill string have been damaged.
- 8- The site supervisor will determine that the equipment are in good condition to resume drilling operations.
- 9- The site supervisor will have the failed rods tagged & returned to the vendor in order to determine the likely cause of the failure.
- 10- The site supervisor will be responsible for completing an incident report that will detail the rod failure event, the recovery operation and resulting corrective actions.

Adds or modifications	Date	Par
Point 2- Add Make sure that the control on neutral position	02-12-2021	S. Tremblay
Adjust the sequence from point #15 to# 29.	06-16-2022	S. Tremblay
Point #25 (former #28) add "cleaning the deflectors and the screens pads before exposing a worker	06-16-2022	S. Tremblay
Add #28 Bit sharpening	06-16-2022	S. Tremblay
Assimilate French and English version so that they are the same	06-16-2022	S. Tremblay
Add photos for example of screen pads	06-16-2022	S. Tremblay
Add point 17) of the possibility that the hydraulic hoses can be stuck and the measures to be taken if this occurs.	12-01-2023	S. Tremblay
Added warning on inspections and high-temperature hoses	7-04-2023	S. Tremblay
Added how to remove a rock from the mast point #26 "Cuttings"	11-10-2023	S. Tremblay D. Baribeault
Addition of a 3-meter distance between the ribbon and the first pieces of equipment in the work area. Addition made to point #8 and on the " STANDARD INSTALLATIONS" following the Amaruk event.	19-12-2023	S. Tremblay S. Desrosiers
Redesign to make English and French procedures the same	23-04-2024	S. Tremblay

STANDARD FOR V-30 BLIND BORING UP 30 IN. DIAMETER

Addition of point 1-b. on the maximum distance of 2 to 4 feet between the table and the collar of the hole. Following actuator failure.	31-10-2024	S. Tremblay D Baribeault JF Gagnon
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Christian St-Amour
Operations Manager

INSTALLATION STANDARD (not scale)

Keep 7 meters between Cubex and booster

A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).

